

Document Control

- System Name: Application
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Executive Summary

- This BRD addresses the ingestion of customer and order CSV data into Delta tables using Delta Live Tables, applying cleansing, enrichment, SCD Type 2 handling, and aggregation to enable high-quality analytical datasets for the Application.

Business Objectives

- Ingest source CSVs for customer and order data into durable Delta tables.
- Cleanse loaded data by removing "Null"/Null literals and duplicate records.
- Enrich order data with a computed TotalAmount field.
- Create an integrated ordersummary table reflecting SCD Type 2 change handling for customer data.
- Aggregate daily spend at the customer level in customeraggregatespend.
- Execute the entire pipeline with Delta Live Tables for enterprise data reliability.

In-Scope

- Ingestion from specific filesystem locations for customer and order data.
- Delta table creation and management for customer, order, ordersummary, and customeraggregatespend.
- Application of cleansing procedures and enrichment logic.
- SCD Type 2 processing within the ordersummary.
- Aggregation and construction of customeraggregatespend.
- Implementation exclusively via Delta Live Tables.

Out-of-Scope

- Any data sources, columns, or systems not explicitly described.
- Security, SLAs, access controls, monitoring, alerting, and performance parameters.
- Scheduling, orchestration, DevOps practices beyond Delta Live Tables pipeline steps.
- User interfaces, downstream analytics, and reporting layers.

Stakeholders

- Information not present in source JSON.

Business Requirements

BR-APP-001 — Source Data Ingestion

- Description: Source customer and order CSVs are to be ingested into Delta tables from defined filesystem locations.
- Business Value: Enables processing on unified, ACID-compliant storage.
- Priority: High

BR-APP-002 — Schema Enforcement

- Description: Customer table columns: CustId, Name, EmailId, Region; Order table columns: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
- Business Value: Promotes data consistency and correct integration.
- Priority: High

BR-APP-003 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables.
- Business Value: Ensures accuracy and reliability of downstream processes.
- Priority: High

BR-APP-004 — Computed TotalAmount

- Description: Add computed column TotalAmount = PricePerUnit × Qty to the order table.
- Business Value: Facilitates downstream aggregation and analytics.
- Priority: Medium

BR-APP-005 — Ordersummary Table Creation

- Description: Create the ordersummary Delta table with SCD Type 2 tracking, joining customer and order on CustId.
- Business Value: Delivers an integrated, historized customer-order dataset.
- Priority: High

BR-APP-006 — SCD Type 2 Handling

- Description: Apply SCD Type 2 logic to ordersummary for customer attribute changes, updating status and timestamps.
- Business Value: Retains auditability and history of customer changes.
- Priority: High

BR-APP-007 — Aggregates for Customer Spend

- Description: Build customeraggregatespend table with total daily spend by customer (group by Name, Date, sum TotalAmount).

- Business Value: Provides key business metrics for spend analysis.
- Priority: Medium

BR-APP-008 — Implementation via Delta Live Tables

- Description: End-to-end pipeline must be orchestrated using Delta Live Tables, adhering to all steps 1–10.
- Business Value: Streamlines deployment, maintenance, and data reliability.
- Priority: High

Assumptions & Constraints

- Requirements follow only the source JSON; no additional assumptions are made.
- Delta Live Tables is the mandated implementation tool.
- Aggregated and SCD Type 2 attributes may require explicit field definitions, which are not detailed in the requirements.

Success Metrics

- 100% of valid records successfully loaded and cleansed.
- All enrichment and business rules applied without data quality exceptions.
- Complete SCD Type 2 history maintained in ordersummary.
- Daily customer spend fully aggregated.
- Zero critical errors in Delta Live Tables execution.

Risks & Mitigations

- Risk: Ambiguity regarding SCD Type 2 attribute field names (StartDate, EndDate, Active).
- Mitigation: Confirm field details with stakeholders before design.
- Risk: Additional required data quality or error handling controls not specified.
- Mitigation: Recommend best practices and stakeholder input during implementation.
- Risk: Lack of environment, scheduling, and security details.
- Mitigation: Flag open questions to business and technical owners.

Approval

- Business Owner: Not present in source JSON
- Technical Lead: Not present in source JSON
- Date: Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Table	Columns	Notes
e/Silver equivalent	customer	CustId, Name, EmailId, Region	Derived from
e/Silver equivalent	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Derived from
Gold equivalent	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Join of cus
	customeraggregatespend	Name, TotalAmount, Date	Aggregate

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Data Cleansing	customer, order	cleansed tables	Remove "Null"/Null, deduplicate
2	Computed Column	order	order (with TotalAmount)	TotalAmount = PricePerUnit × Qty
3	Join and SCD Type 2	customer + order	ordersummary	Join on CustId, SCD Type 2 change handling for customer
4	Aggregation	ordersummary	customeraggregatespend	Group by Name, Date, sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computed Column	TotalAmount = PricePerUnit × Qty	Requirements input
BR-2	Cleansing	Remove "Null"/Null, deduplication	Requirements input
BR-3	Join	Join customer to order on CustId	Requirements input
BR-4	SCD Type 2	Set previous recs to Inactive, new to Active, maintain history	Requirements input
BR-5	Aggregation	Group by Name, Date, sum(TotalAmount)	Requirements input

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order data from CSV source paths	None
2	Cleanse data and add computed columns	Step 1
3	Join and apply SCD Type 2 logic in ordersummary	Step 2
4	Aggregate spend to generate customeraggregatespend	Step 3

Appendix F: Process Flow Diagram

Figure 1: End-to-End Data Pipeline Flow

Start



End